

Breast cancer risk and hysterectomy status: the Multiethnic Cohort study

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Abstract

Objective The main objective was to examine the association between simple hysterectomy (without bilateral oophorectomy) and breast cancer risk. Because hysterectomy prevalence varies by ethnicity, the secondary objective was to examine whether inclusion of women with hysterectomies affects the estimates of breast cancer risk by ethnicity.

Methods The Multiethnic Cohort study was assembled between 1993 and 1996 and included 68,065 women from Hawaii and Los Angeles, aged 45–75 years, without any missing information or bilateral oophorectomy. Hysterectomy status was self-reported. After 7.7 years median follow-up, 1,862 cases of invasive breast cancer were identified. Proportional hazards models were used to estimate relative risks (RR) while controlling for known risk factors.

Results Prevalence of simple hysterectomy varied from 12% to 29% among the ethnic groups (White, African American, Native Hawaiian, Japanese American, and Latina). Overall, hysterectomy was not associated with breast cancer risk (RR = 0.98). Although the RRs were nonsignificantly elevated by 15% in White women and nonsignificantly reduced by 15% in Latinas of non-US origin, the variation by ethnicity was not significant

($p_{\text{interaction}} = 0.48$). The breast cancer risk associated with ethnicity was very similar when estimated with and without women with hysterectomies.

Conclusions This study suggests that simple hysterectomy status does not alter breast cancer risk. Therefore, inclusion of women with simple hysterectomies does not substantially change estimated risk of breast cancer by ethnicity.

Keywords Breast cancer · Hysterectomy · Ethnicity · Cohort

Introduction

It is important to investigate ways to identify women at higher risk for breast cancer as a means to understand the biology of the disease and to determine who may be most appropriate for intensified screening or preventive interventions. For example, clinical factors identified by way of routine examinations and medical history, such as bone mineral density and bilateral oophorectomy status, have been examined for their associations with breast cancer risk. In this regard, a few retrospective cohort studies [1, 2] and case-control studies [3, 4] have examined the association between simple hysterectomy (i.e., without bilateral oophorectomy) and breast cancer risk. The results from these studies have been inconsistent, with one showing a 30% increase in risk [1] and the others showing reductions in risk from 13% to 30% [2–4]. The crude odds ratio for hysterectomy was also estimated from data presented in a large pooled analysis examining hormone therapy use and breast cancer risk as 0.90 (95% CI = 0.87–0.93) [5]. Because hysterectomy is very common and risk factors for conditions leading to hysterectomy may also increase

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breast cancer risk, our primary objective was to investigate this association within the prospective Multiethnic Cohort (MEC) study.

In 2002, Pike et al. [6] examined the incidence of breast cancer in the MEC and found that a number of ethnic differences in incidence were not explained by known breast cancer risk factors. As compared to White women, Latinas not born in the United States were at nonsignificantly reduced risk of breast cancer, whereas Native Hawaiian women were at significantly increased risk after multivariate adjustment. Women who had a simple hysterectomy before menopause were excluded from this analysis because imputation of the unknown age at menopause has the potential to bias the associations of age at menopause and hormone therapy use with breast cancer risk, and thus lead to incomplete or biased adjustment when examining whether or not these factors explained the differences in breast cancer incidence by ethnicity [7, 8]. Because the prevalence rates of hysterectomy differ among ethnic groups, with African American women undergoing a hysterectomy substantially more often than White women [9–14], and because hysterectomy may be associated with breast cancer risk, inclusion of women who had a simple hysterectomy may lead to different incidence rates. Thus, a secondary objective of the present analysis was to examine ethnic differences in breast cancer risk including and excluding women with simple hysterectomies.

Materials and methods

The MEC was designed to investigate lifestyle and dietary factors with respect to cancer outcomes in five different ethnic groups in Hawaii and Los Angeles aged 45–75 years at baseline; details about the study have been published previously [15]. The study was approved by the institutional review boards at the University of Hawaii and the University of Southern California. The cohort was assembled between 1993 and 1996 using drivers' license records in both states, supplemented with voter registration lists in Hawaii and Health Care Financing Administration files in Los Angeles. A self-administered questionnaire was completed and returned by mail by about 215,000 people of whom 118,869 were women. Excluded from the present analysis were 6,654 women with previous self-reported or cancer registry-detected diagnoses of breast, endometrial, or ovarian cancer, 8,051 women who did not belong to one of the five targeted ethnic groups (White/Caucasian, African American, Native Hawaiian, Japanese American, Latina), 11,810 women who had bilateral oophorectomy, and 4,611 women who had implausible responses on the dietary portion of the questionnaire [16]. Women ($n = 19,678$) with missing values in the exposure or

covariate variables, including extreme values for height or weight that resulted in a body mass index (BMI) outside of the 15–50 kg/m² range, were excluded. Thus, 68,065 women were included in this analysis.

Exposure and covariate information from baseline questionnaire

The baseline questionnaire data included information about demographic factors, diet, reproductive and menstrual history, and family history of cancer. Women who reported that their menstrual periods stopped due to surgery, and responded “Yes” in response to the question “Have you ever had a hysterectomy (complete removal of the uterus)?” but did not respond, “Yes, both ovaries” to the question “Have you ever had your ovaries removed?” were classified as having had a simple hysterectomy; thus, women classified in our exposed group are women who had a simple hysterectomy before menopause. Age at hysterectomy was determined for these women by a question inquiring how old subjects were when their menstrual periods stopped permanently. Women without simple hysterectomy (unexposed) were those who were premenopausal or experienced menopause naturally without surgery. Women with bilateral oophorectomy were excluded because of their reduced risk for breast cancer. Ethnicity was based on self-report; women reporting mixed ancestry were assigned to a single group based on the priority ranking: African American, Native Hawaiian, Latina, Japanese American, and White [15]. For the present analysis, Latina women were divided into two groups: those who were born in the United States and those who were not born in the United States but were born in Mexico, Central, or South America.

Follow-up and identification of cases

Eligibility for cohort inclusion was to be 45–75 years of age but a few subjects (<1%) returned their questionnaires at ages 42–44. Thus, follow-up time was initiated at the date of return of the baseline questionnaire or the 45th birthday of the subject, whichever was last, and was accrued until the date of diagnosis of breast, endometrial, or ovarian cancer, death, or the last follow-up date (31 December 2002). Incident diagnoses of cancers were identified by linkage with the Los Angeles County Cancer Surveillance Program, the State of California Cancer Registry, and the Hawaii Tumor Registry, which together cover the entire population of the two states. These registries are members of the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program. Case ascertainment was complete to the end of 2002. Deaths were identified by linkage with the databases of the Hawaii and California

vital statistics offices and the National Death Index. The primary analyses included women with invasive breast cancers as cases. The median follow-up time for women included in this analysis was 7.7 years (interquartile range: 7.2–8.3 years), during which time 1,862 cases of invasive cancer were identified. Information about estrogen and progesterone receptor status, stage, and histology was derived from SEER registry data.

Statistical analysis

Women with a simple hysterectomy were compared to women without a simple hysterectomy with respect to breast cancer risk factors. Logistic regression modeling with hysterectomy status as the outcome was done to test if the odds of having a simple hysterectomy were different by the levels of the risk factors adjusted for age and ethnicity.

Cox proportional hazards models [17], with age in days as the time metric, were used to estimate the relative risks (RR) of breast cancer associated with having had a simple hysterectomy. Because we could not determine the menopausal status of the women with a simple hysterectomy, both pre- and postmenopausal women were included in the main analysis. All models were adjusted for ethnicity and then separate models were done for each ethnic group. All models were stratified by follow-up time (≤ 2 , >2 – 5 , >5 years). Additional models were adjusted for education (≤ 12 th grade, >12 th grade), age at menarche (≤ 12 , 13–14, ≥ 15 years), age at first birth (never, ≤ 20 , 21–30, >30 years), number of children (0–1, 2–3, ≥ 4), duration of current estrogen with progestin use (continuous), duration of current estrogen only use (continuous), duration of past estrogen with progestin use (continuous), BMI (<25 , 25 – <30 , ≥ 30 kg/m²), alcohol consumption (none, ≤ 1 , >1 – <2 , ≥ 2 drinks/day), and family history of breast cancer in a mother or sister (no, yes). We found that adjustment by history of mammography screening (within two years, more than 2 years, never) did not change the results and, thus, did not include this variable as a confounder. The significance of cross-product terms between ethnicity and hysterectomy status was tested with a Wald test to examine whether the associations differed among the ethnic groups. The proportionality assumption of the hazards models was tested by examining the Kaplan–Meier curves and assessing the Schoenfeld residuals.

Supplementary analyses were done to examine the effect of hysterectomy on subgroups of breast cancer as defined by estrogen receptor (ER) status, progesterone receptor (PR) status, joint ER/PR status, stage (in situ, localized, advanced), and histology (infiltrating ductal carcinoma, ICD-O2 8500; lobular and mixed ductal lobular, ICD-O2 8520, 8522; mucinous, ICD-O2 8480). The interaction between hysterectomy status and history of mammography

screening (within 2 years, more than 2 years, never) was also examined. Because we could not determine menopausal status in women who had a simple hysterectomy, we did a supplementary analysis restricted to women 60 years of age or older at baseline, all of whom would likely be postmenopausal, to get an indication of whether or not the effect may be modified by menopausal status.

Finally, we used Cox proportional hazards models to examine the association between ethnicity and breast cancer risk. We compared the results in all women and a subset which excluded women with simple hysterectomies. All analyses were done using SAS version 9.1.

Results

The prevalence of simple hysterectomy in cohort members with intact ovaries was 19%, but the prevalence differed by ethnicity. The prevalence was highest in African American women ($n = 3,570$, 29%) and Latinas born in the US ($n = 1,774$, 24%) whereas it was lowest in Japanese American women ($n = 2,331$, 12%). Women in the other ethnic groups had a moderate prevalence of hysterectomy (16–18%). Compared with women without hysterectomies, women who had simple hysterectomies, on average, were older and a higher proportion had an earlier menarche, an earlier age at first birth, a greater number of children, a BMI in the overweight or obese range, 12 years of education or less, and a history of breast cancer in a first degree relative (Table 1). They were also slightly more likely to have had a mammogram and to be using estrogen only hormonal therapy at baseline.

Hysterectomy status was not associated with the risk of breast cancer overall (RR = 0.98; 95% CI = 0.86–1.11), or in any of the ethnic groups examined (Table 2). The RR in White women was nonsignificantly elevated (1.15; 95% CI = 0.89–1.47) and the RR in Latinas of non-US origin was nonsignificantly reduced (0.85; 95% CI = 0.48–1.51), but the RRs did not vary significantly across the ethnic groups examined ($p_{\text{interaction}} = 0.48$). We did not find an association with age at hysterectomy. The overall association was similar when the analysis was restricted to women who were 60 years of age or older at cohort entry (RR = 1.05; 95% CI = 0.88–1.24) and when the analysis was done by history of mammography screening (within 2 years, more than 2 years, never; $p_{\text{interaction}} = 0.90$). Furthermore, hysterectomy status was not associated with the subgroups of breast cancer defined by ER status, PR status, stage, or histology. The risk of carcinoma in situ also was not associated with hysterectomy status (RR = 0.86; 95% CI = 0.65–1.14).

With respect to our secondary objective, we observed that the risks of breast cancer according to ethnicity were

Table 1 Baseline characteristics of the women in the Multiethnic Cohort, by hysterectomy status

Characteristic	No simple hysterectomy ^a		Simple hysterectomy		<i>p</i> ^b
	<i>n</i>	(%)	<i>n</i>	(%)	
Age (years)					
<60	30327	(54.9)	6004	(47.0)	<0.0001
≥60	24953	(45.1)	6781	(53.0)	
Ethnicity					
White	14289	(25.8)	3207	(25.1)	<0.0001
African American	8604	(15.6)	3570	(27.9)	
Native Hawaiian	4233	(7.7)	830	(6.5)	
Japanese American	16912	(30.6)	2331	(18.2)	
Latina–US born	5692	(10.3)	1774	(13.9)	
Latina–not US born	5550	(10.0)	1073	(8.4)	
Age at menopause or hysterectomy (years)					
Premenopausal	13248	(24.0)	–		–
<45	6620	(12.0)	9761	(76.3)	
45–49	13312	(24.1)	2202	(17.2)	
≥50	22100	(40.0)	822	(6.4)	
Age at menarche (years)					
≤12	26948	(48.7)	6798	(53.2)	<0.0001
13–14	21561	(39.0)	4680	(36.6)	
≥15	6771	(12.2)	1307	(10.2)	
Age at first birth (years)					
Never	7193	(13.0)	1306	(10.2)	<0.0001
≤20	14193	(25.7)	5021	(39.3)	
21–30	29434	(53.2)	6027	(47.1)	
≥30	4460	(8.1)	431	(3.4)	
Number of children					
0	7193	(13.0)	1306	(10.2)	<0.0001
1	6272	(11.3)	1299	(10.2)	
2–3	25141	(45.5)	5512	(43.1)	
≥4	16674	(30.2)	4668	(36.5)	
BMI (kg/m ²)					
<25	29490	(53.3)	5569	(43.6)	<0.0001
25–<30	16012	(29.0)	4306	(33.7)	
≥30	9778	(17.7)	2910	(22.8)	
Current estrogen and progestin use					
No	45701	(82.7)	12478	(97.6)	<0.0001
Yes	9579	(17.3)	307	(2.4)	
Current estrogen use					
No	52807	(95.5)	8369	(65.5)	<0.0001
Yes	2473	(4.5)	4416	(34.5)	
Past estrogen and progestin use					
No	51497	(93.2)	11831	(92.5)	<0.0001
Yes	3783	(6.8)	954	(7.5)	
Alcohol intake (drinks/day)					
Never	41869	(75.7)	9734	(76.1)	0.0003
<1	8275	(15.0)	1922	(15.0)	
1–<2	2678	(4.8)	629	(4.9)	
≥2	2458	(4.4)	500	(3.9)	

Table 1 continued

Characteristic	No simple hysterectomy ^a		Simple hysterectomy		<i>p</i> ^b
	<i>n</i>	(%)	<i>n</i>	(%)	
Education (years)					
≤12	23357	(42.3)	5957	(46.6)	<0.0001
>12	31923	(57.7)	6828	(53.4)	
Family history ^c					
No	49452	(89.5)	11244	(87.9)	0.0006
Yes	5828	(10.5)	1541	(12.1)	
Previous mammography					
Never	7153	(12.9)	1044	(8.2)	<0.0001
>2 years ago	6020	(10.9)	1347	(10.5)	
Within 2 years	42107	(76.2)	10394	(81.3)	

^a Premenopausal women and postmenopausal women who had natural menopause

^b *p*-value from logistic regression adjusted for age (continuous) and ethnicity. Age is adjusted for ethnicity only and ethnicity is adjusted for age only

^c Family history of breast cancer in mother or a sister

Table 2 Multivariate adjusted relative risks (RR) of breast cancer according to hysterectomy status, by ethnicity

Ethnicity	Simple hysterectomy	Cases	Age adjusted		Fully adjusted ^a	
			RR	(95% CI)	RR	(95% CI)
White	No	390	1.0		1.0	
	Yes	101	1.07	(0.86–1.33)	1.15	(0.89–1.47)
African American	No	232	1.0		1.0	
	Yes	88	0.90	(0.70–1.15)	0.90	(0.69–1.16)
Native Hawaiian	No	153	1.0		1.0	
	Yes	29	0.85	(0.57–1.27)	0.89	(0.58–1.35)
Japanese American	No	527	1.0		1.0	
	Yes	69	0.92	(0.71–1.18)	0.89	(0.67–1.19)
Latina–US born	No	126	1.0		1.0	
	Yes	43	1.05	(0.74–1.48)	1.10	(0.76–1.60)
Latina–not US born	No	90	1.0		1.0	
	Yes	14	0.80	(0.46–1.41)	0.85	(0.48–1.51)
Overall ^b	No	1518	1.0		1.0	
	Yes	344	0.93	(0.83–1.05)	0.98	(0.86–1.11)
Overall, by age at hysterectomy	No	1518	1.0		1.0	
	Yes, at <45 years of age	250	0.90	(0.78–1.03)	0.94	(0.81–1.09)
	Yes, at 45–49 years of age	69	1.05	(0.82–1.33)	1.10	(0.86–1.41)
	Yes, at 50+ years of age	25	0.99	(0.66–1.47)	1.03	(0.69–1.54)

^a Adjusted for age, BMI, family history in a mother or sister, education, alcohol consumption, age at menarche, age at first birth, number of children, duration of current estrogen with progestin use, duration of current estrogen only use, and duration of past estrogen with progestin use. For analyses including all women, additionally adjusted for ethnicity

^b $p_{\text{interaction}} = 0.48$ between hysterectomy (no, yes) and ethnicity when added to this model

very similar whether estimated with or without women with simple hysterectomies (Table 3). As compared to White women, African American women and Latinas born in the US had similar risk for breast cancer. At increased risk for breast cancer were Native Hawaiian women

(including women with simple hysterectomy, RR = 1.72; 95% CI = 1.44–2.06) and Japanese American women (including women with simple hysterectomy, RR = 1.19; 95% CI = 1.04–1.35). At reduced risk for breast cancer were Latinas not born in the US, but this only reached

Table 3 Multivariate adjusted relative risks (RR) of breast cancer according to ethnicity, comparing different groups defined by hysterectomy status

Ethnicity	All women ^a			Excluding women with simple hysterectomy		
	Cases	RR ^b	(95% CI)	Cases	RR ^b	(95% CI)
White	491	1.0		390	1.0	
African American	320	1.01	(0.86–1.16)	232	1.07	(0.90–1.27)
Native Hawaiian	182	1.72	(1.44–2.06)	153	1.80	(1.48–2.20)
Japanese American	596	1.19	(1.04–1.35)	527	1.25	(1.08–1.44)
Latina–US born	169	0.92	(0.77–1.10)	125	0.94	(0.76–1.16)
Latina–not US born	104	0.77	(0.61–0.96)	90	0.83	(0.65–1.06)

^a Premenopausal women, women with natural menopause, and women with simple hysterectomy

^b Adjusted for age, BMI, family history in a mother or sister, education, alcohol consumption, age at menarche, age at first birth, number of children, duration of current estrogen with progestin use, duration of current estrogen only use, and duration of past estrogen with progestin use

statistical significance in the larger sample that included women with simple hysterectomies (RR = 0.77; 95% CI = 0.61–0.96). Because adjustment for menopausal status and age at menopause cannot be done without imputation when including women with simple hysterectomies, we investigated the confounding effect in the sample excluding these women. The change in the estimated RRs for the association between ethnicity and breast cancer risk after adjustment for age at menopause was less extreme than -2.8% [$100\% \times (RR_{\text{not adjusted}} - RR_{\text{adjusted}}) / RR_{\text{not adjusted}}$] [18]. Thus, it is not likely that the risk estimates shown in Table 3 are strongly confounded by menopausal status and age at menopause.

Discussion

As compared to other studies, we found neither an increased [1] nor a decreased [2–4] risk of breast cancer in relation to simple hysterectomy status. In agreement with other studies, we did not find an association with age at surgery [1–3, 5]. Given these null findings, inclusion of women with hysterectomy in the analysis of the association between ethnicity and breast cancer risk did not materially change the conclusions about this association despite large variation in hysterectomy prevalence by ethnic group.

The balance of several phenomena, including measurement error and biological mechanisms that may be weak or counteract one another, could have resulted in the null association that was estimated between hysterectomy and breast cancer risk in this study. A biological mechanism that could increase risk is that conditions leading to hysterectomy may share a hormonal etiology with breast cancer. Indications for hysterectomy include leiomyomas (uterine fibroids), endometriosis, uterine prolapse, and menstrual disorders [12, 19–21]. Risk factors for some of these conditions are also known breast cancer risk factors:

early age at menarche, low parity, BMI, and low physical activity [10, 22–29]. There are exceptions. Alcohol intake and BMI decrease the risk for endometriosis [30–32] but increase the risk for postmenopausal breast cancer. Parity and early age at first birth increase the risk of prolapse [33–35] but decrease the risk of breast cancer. Further, the conditions themselves should be associated with breast cancer but studies of endometriosis show no overall association [1, 36–39] or a slightly increased risk [40] and the one study of hysterectomy for leiomyomas reported only a slight increased risk [1].

Another biological mechanism, but which could reduce the risk associated with hysterectomy, is that simple hysterectomy seems to reduce blood flow to the ovaries [41] and, consequently, lessens the production of estrogens and hastens menopause. Simple hysterectomy is associated with changes in ovarian histology [42], differential expression of menopausal symptoms [43–45], and increases in follicle-stimulating hormone [46]. Women with a simple hysterectomy have an average age at menopause as determined by follicle-stimulating hormone levels >40 IU/L that is 3.7 years younger than women who have a natural menopause [47]. Given that each year of earlier age at menopause reduces breast cancer risk by 2–3% [5, 48], 3.7 years earlier would translate into a risk reduction of 7–11%. The reduction in risk could be larger because hysterectomy may have effects on hormone levels even after menopause; levels of testosterone and estrone are slightly lower in women who had a simple hysterectomy than women who had a natural menopause [49].

The null association that was estimated between hysterectomy and breast cancer risk could also be influenced by the possibility that women may have taken pharmacological treatments for the indications for which they had a hysterectomy prior to having the surgery and these treatments could have affected breast cancer risk. For example, treatments given for leiomyomas and endometriosis, two of

the main indications for hysterectomy, may include danazol (an androgenic medication), oral contraceptives, and gonatropin releasing hormone agonists [50–52].

Finally, possible measurement error in hysterectomy status could influence the association that was estimated as null. Although one study showed excellent concordance between hospital records and self-reported hysterectomy and oophorectomy [53], other studies have shown less concordance between self-report and information from physicians [54], hospital records [55], a second self-report up to 10 years later [59], or a second self-report within a short period of time [56, 57]. The study with excellent concordance was among nurses, who are likely more medically informed than the general population [53]. Because bilateral oophorectomy reduces breast cancer risk (in the current study, relative to women having natural menopause, $RR = 0.81$; 95% $CI = 0.69$ – 0.94), if women with bilateral oophorectomy were included in the simple hysterectomy group, the risk associated with hysterectomy would be biased toward a reduction in risk. This bias may be more apparent in studies using self-report than in studies using medical records. Although slightly reduced risks were observed in two case–control studies using self-reported data [3, 4], reduced risks were also observed in one of the two retrospective studies using medical records [2]. Conversely, women who were premenopausal at the time of entry into the cohort study could subsequently have had a simple hysterectomy in the follow-up time. The person-years incorrectly classified should be minimal, however, because few women were premenopausal at baseline and those women who were premenopausal were over 45 years of age.

The balance of these biological mechanisms and potential biases could be different among different populations such that some studies estimate an increased risk, some estimate a decreased risk, and some show no risk difference with hysterectomy. On the other hand, no real association may exist between hysterectomy and breast cancer risk.

As in other studies [12, 21, 58], we found that the prevalence of simple hysterectomy differed by ethnicity; it was relatively high in African American women and low in Japanese American women. This difference in prevalence could be due to ethnic differences in the incidence and severity of uterine pathologies, prevalence of early treatment to prevent hysterectomy, or medical practice [12, 59]. We had been concerned that if hysterectomy was associated with breast cancer risk and also associated with ethnicity, then the exclusion of women with simple hysterectomies would affect the association between ethnicity and breast cancer risk. Their exclusion, however, did not substantially change the association and supports the generalizability of this observation. Furthermore, the association was

remarkably similar to results presented previously for the MEC [6] despite 2–3 years additional follow-up and a sample with premenopausal women, women with a simple hysterectomy, and naturally postmenopausal women as compared to using a sample with naturally postmenopausal women and women with bilateral oophorectomy.

Although we did not have specific data to disentangle the effects of differing age at menopause, indication for hysterectomy, or possible misreporting of hysterectomy and oophorectomy status, this study and others have not shown an increased risk of breast cancer associated with simple hysterectomy. Despite the considerable variation in hysterectomy prevalence by ethnicity, because of the lack of association between hysterectomy and breast cancer risk, exclusion of women with simple hysterectomies did not substantially change the observed differences in breast cancer risk by ethnicity.

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References

1. Schairer C, Persson I, Falkeborn M et al (1997) Breast cancer risk associated with gynecologic surgery and indications for such surgery. *Int J Cancer* 2:150–154. doi:10.1002/(SICI)1097-0215(19970117)70:2<150::AID-IJC2>3.0.CO;2-W
2. Kreiger N, Sloan M, Cotterchio M, Kirsh V (1999) The risk of breast cancer following reproductive surgery. *Eur J Cancer* 1:97–101. doi:10.1016/S0959-8049(98)00343-8
3. Irwin KL, Lee NC, Peterson HB et al (1988) Hysterectomy, tubal sterilization, and the risk of breast cancer. *Am J Epidemiol* 6:1192–1201
4. Parazzini F, Braga C, La VC et al (1997) Hysterectomy, oophorectomy in premenopause, and risk of breast cancer. *Obstet Gynecol* 3:453–456. doi:10.1016/S0029-7844(97)00295-0
5. Collaborative Group on Hormonal Factors in Breast Cancer (1997) Breast cancer and hormone replacement therapy: collaborative reanalysis of data from 51 epidemiological studies of 52,705 women with breast cancer and 108,411 women without breast cancer. Collaborative Group on Hormonal Factors in Breast Cancer. *Lancet* 9084:1047–1059
6. Pike MC, Kolonel LN, Henderson BE et al (2002) Breast cancer in a multiethnic cohort in Hawaii and Los Angeles: risk factor-adjusted incidence in Japanese equals and in Hawaiians exceeds that in whites. *Cancer Epidemiol Biomarkers Prev* 9:795–800
7. Pike MC, Ross RK, Spicer DV (1998) Problems involved in including women with simple hysterectomy in epidemiologic studies measuring the effects of hormone replacement therapy on breast cancer risk. *Am J Epidemiol* 8:718–721
8. Rockhill B, Colditz GA, Rosner B (2000) Bias in breast cancer analyses due to error in age at menopause. *Am J Epidemiol* 4:404–408
9. Powell LH, Meyer P, Weiss G et al (2005) Ethnic differences in past hysterectomy for benign conditions. *Womens Health Issues* 4:179–186. doi:10.1016/j.whi.2005.05.002

10. Marshall LM, Spiegelman D, Goldman MB et al (1998) A prospective study of reproductive factors and oral contraceptive use in relation to the risk of uterine leiomyomata. *Fertil Steril* 3:432–439. doi:[10.1016/S0015-0282\(98\)00208-8](https://doi.org/10.1016/S0015-0282(98)00208-8)
11. Brett KM, Marsh JV, Madans JH (1997) Epidemiology of hysterectomy in the United States: demographic and reproductive factors in a nationally representative sample. *J Womens Health* 3:309–316
12. Meilahn EN, Matthews KA, Egeland G, Kelsey SF (1989) Characteristics of women with hysterectomy. *Maturitas* 4:319–329. doi:[10.1016/0378-5122\(89\)90028-5](https://doi.org/10.1016/0378-5122(89)90028-5)
13. Sherman ME, Carreon JD, Lacey JV Jr, Devesa SS (2005) Impact of hysterectomy on endometrial carcinoma rates in the United States. *J Natl Cancer Inst* 22:1700–1702
14. Chlebowski RT, Chen Z, Anderson GL et al (2005) Ethnicity and breast cancer: factors influencing differences in incidence and outcome. *J Natl Cancer Inst* 6:439–448
15. Kolonel LN, Henderson BE, Hankin JH et al (2000) A Multiethnic Cohort in Hawaii and Los Angeles: baseline characteristics. *Am J Epidemiol* 151:346–357
16. Nothlings U, Wilkens LR, Murphy SP et al (2005) Meat and fat intake as risk factors for pancreatic cancer: the Multiethnic Cohort study. *J Natl Cancer Inst* 19:1458–1465
17. Therneau T, Grambsch P (2000) Modeling survival data: extending the Cox model. Springer-Verlag, New York
18. Rothman KJ, Greenland S (1998) Modern epidemiology, 2nd edn. Lippincott-Raven, Philadelphia, PA, p 256
19. Farquhar CM, Steiner CA (2002) Hysterectomy rates in the United States 1990–1997. *Obstet Gynecol* 2:229–234. doi:[10.1016/S0029-7844\(01\)01723-9](https://doi.org/10.1016/S0029-7844(01)01723-9)
20. Jacobson GF, Shaber RE, Armstrong MA, Hung YY (2006) Hysterectomy rates for benign indications. *Obstet Gynecol* 6:1278–1283
21. Keshavarz H, Hillis SD, Kieke BA, Marchbanks PA (2002) Hysterectomy surveillance – United States, 1994–1999. *MMWR CDC Surveill Summ* 51:1–8
22. Parazzini F (2006) Risk factors for clinically diagnosed uterine fibroids in women around menopause. *Maturitas* 2:174–179. doi:[10.1016/j.maturitas.2006.01.013](https://doi.org/10.1016/j.maturitas.2006.01.013)
23. Settles A, Andreassen AH, Jorgensen T (2005) Hypertension is associated with an increased risk for hysterectomy: a Danish cohort study. *Eur J Obstet Gynecol Reprod Biol* 2:218–224. doi:[10.1016/j.ejogrb.2005.02.010](https://doi.org/10.1016/j.ejogrb.2005.02.010)
24. Terry KL, De Vivo I, Hankinson SE et al (2007) Anthropometric characteristics and risk of uterine leiomyoma. *Epidemiology* 6:758–763
25. Wise LA, Palmer JR, Spiegelman D et al (2005) Influence of body size and body fat distribution on risk of uterine leiomyomata in U.S. black women. *Epidemiology* 3:346–354. doi:[10.1097/01.ede.0000158742.11877.99](https://doi.org/10.1097/01.ede.0000158742.11877.99)
26. Wise LA, Palmer JR, Harlow BL et al (2004) Reproductive factors, hormonal contraception, and risk of uterine leiomyomata in African-American women: a prospective study. *Am J Epidemiol* 2:113–123. doi:[10.1093/aje/kwh016](https://doi.org/10.1093/aje/kwh016)
27. Farquhar C (2007) Endometriosis. *BMJ* 7587:249–253. doi:[10.1136/bmj.39073.736829.BE](https://doi.org/10.1136/bmj.39073.736829.BE)
28. Heilner JF, Donnez J, Nackers F et al (2007) Environmental and host-associated risk factors in endometriosis and deep endometriotic nodules: a matched case-control study. *Environ Res* 1:121–129. doi:[10.1016/j.envres.2006.04.004](https://doi.org/10.1016/j.envres.2006.04.004)
29. Missmer SA, Hankinson SE, Spiegelman D et al (2004) Reproductive history and endometriosis among premenopausal women. *Obstet Gynecol* 5(Pt 1):965–974
30. Missmer SA, Hankinson SE, Spiegelman D et al (2004) Incidence of laparoscopically confirmed endometriosis by demographic, anthropometric, and lifestyle factors. *Am J Epidemiol* 8:784–796. doi:[10.1093/aje/kwh275](https://doi.org/10.1093/aje/kwh275)
31. Parazzini F, Chiaffarino F, Surace M et al (2004) Selected food intake and risk of endometriosis. *Hum Reprod* 8:1755–1759. doi:[10.1093/humrep/deh395](https://doi.org/10.1093/humrep/deh395)
32. Vigano P, Parazzini F, Somigliana E, Vercellini P (2004) Endometriosis: epidemiology and aetiological factors. *Best Pract Res Clin Obstet Gynaecol* 2:177–200. doi:[10.1016/j.bpobgyn.2004.01.007](https://doi.org/10.1016/j.bpobgyn.2004.01.007)
33. Nygaard I, Bradley C, Brandt D (2004) Pelvic organ prolapse in older women: prevalence and risk factors. *Obstet Gynecol* 3:489–497
34. Rortveit G, Brown JS, Thom DH et al (2007) Symptomatic pelvic organ prolapse: prevalence and risk factors in a population-based, racially diverse cohort. *Obstet Gynecol* 6:1396–1403
35. Tegerstedt G, Miedel A, Maehle-Schmidt M, Nyren O, Hammarstrom M (2006) Obstetric risk factors for symptomatic prolapse: a population-based approach. *Am J Obstet Gynecol* 1:75–81. doi:[10.1016/j.ajog.2005.06.086](https://doi.org/10.1016/j.ajog.2005.06.086)
36. Bertelsen L, Mellekjaer L, Frederiksen K et al (2007) Risk for breast cancer among women with endometriosis. *Int J Cancer* 6:1372–1375. doi:[10.1002/ijc.22490](https://doi.org/10.1002/ijc.22490)
37. Brinton LA, Westhoff CL, Scoccia B et al (2005) Causes of infertility as predictors of subsequent cancer risk. *Epidemiology* 4:500–507. doi:[10.1097/01.ede.0000164812.02181.d5](https://doi.org/10.1097/01.ede.0000164812.02181.d5)
38. Olson JE, Cerhan JR, Janney CA et al (2002) Postmenopausal cancer risk after self-reported endometriosis diagnosis in the Iowa Women's Health Study. *Cancer* 5:1612–1618. doi:[10.1002/cncr.10370](https://doi.org/10.1002/cncr.10370)
39. Melin A, Sørensen P, Bergqvist A (2007) The risk of cancer and the role of parity among women with endometriosis. *Hum Reprod* 11:3021–3026. doi:[10.1093/humrep/dem209](https://doi.org/10.1093/humrep/dem209)
40. Brinton LA, Gridley G, Persson I, Baron J, Bergqvist A (1997) Cancer risk after a hospital discharge diagnosis of endometriosis. *Am J Obstet Gynecol* 3:572–579. doi:[10.1016/S0002-9378\(97\)70550-7](https://doi.org/10.1016/S0002-9378(97)70550-7)
41. Chan CC, Ng EH, Ho PC (2005) Ovarian changes after abdominal hysterectomy for benign conditions. *J Soc Gynecol Investig* 1:54–57. doi:[10.1016/j.jsgi.2004.07.004](https://doi.org/10.1016/j.jsgi.2004.07.004)
42. Souza AZ, Fonseca AM, Izzo VM, Clauzet RM, Salvatore CA (1986) Ovarian histology and function after total abdominal hysterectomy. *Obstet Gynecol* 6:847–849
43. Oldenhave A, Jazmann LJ, Everaerd WT, Haspels AA (1993) Hysterectomized women with ovarian conservation report more severe climacteric complaints than do normal climacteric women of similar age. *Am J Obstet Gynecol* 3(Pt 1):765–771
44. Hartmann BW, Kirchengast S, Albrecht A, Metka M, Huber JC (1995) Hysterectomy increases the symptomatology of postmenopausal syndrome. *Gynecol Endocrinol* 3:247–252
45. Stadberg E, Mattsson LA, Milsom I (2000) Factors associated with climacteric symptoms and the use of hormone replacement therapy. *Acta Obstet Gynecol Scand* 4:286–292. doi:[10.1034/j.1600-0412.2000.079004286.x](https://doi.org/10.1034/j.1600-0412.2000.079004286.x)
46. Cooper GS, Thorp JM Jr (1999) FSH levels in relation to hysterectomy and to unilateral oophorectomy. *Obstet Gynecol* 6:969–972. doi:[10.1016/S0029-7844\(99\)00429-9](https://doi.org/10.1016/S0029-7844(99)00429-9)
47. Farquhar CM, Sadler L, Harvey SA, Stewart AW (2005) The association of hysterectomy and menopause: a prospective cohort study. *BJOG* 7:956–962. doi:[10.1111/j.1471-0528.2005.00696.x](https://doi.org/10.1111/j.1471-0528.2005.00696.x)
48. Rosner B, Colditz GA (1996) Nurses' health study: log-incidence mathematical model of breast cancer incidence. *J Natl Cancer Inst* 6:359–364. doi:[10.1093/jnci/88.6.359](https://doi.org/10.1093/jnci/88.6.359)
49. Setiawan VW, Haiman CA, Stanczyk FZ, Le ML, Henderson BE (2006) Racial/ethnic differences in postmenopausal endogenous hormones: the Multiethnic Cohort study. *Cancer Epidemiol*

- Biomarkers Prev 10:1849–1855. doi:[10.1158/1055-9965.EPI-06-0307](https://doi.org/10.1158/1055-9965.EPI-06-0307)
50. Davis L, Kennedy SS, Moore J, Prentice A (2007) Modern combined oral contraceptives for pain associated with endometriosis. *Cochrane Database Syst Rev* 3:CD001019
 51. Parker WH (2007) Uterine myomas: management. *Fertil Steril* 2:255–271. doi:[10.1016/j.fertnstert.2007.06.044](https://doi.org/10.1016/j.fertnstert.2007.06.044)
 52. Selak V, Farquhar C, Prentice A, Singla A (2007) Danazol for pelvic pain associated with endometriosis. *Cochrane Database Syst Rev* 4:CD000068
 53. Colditz GA, Stampfer MJ, Willett WC et al (1987) Reproducibility and validity of self-reported menopausal status in a prospective cohort study. *Am J Epidemiol* 126:319–325
 54. Treloar SA, Do KA, O'Connor VM et al (1999) Predictors of hysterectomy: an Australian study. *Am J Obstet Gynecol* 4:945–954. doi:[10.1016/S0002-9378\(99\)70666-6](https://doi.org/10.1016/S0002-9378(99)70666-6)
 55. Brett KM, Madans JH (1994) Hysterectomy use: the correspondence between self-reports and hospital records. *Am J Public Health* 10:1653–1655
 56. Bosetti C, Tavani A, Negri E, Trichopoulos D, La VC (2001) Reliability of data on medical conditions, menstrual and reproductive history provided by hospital controls. *J Clin Epidemiol* 9:902–906. doi:[10.1016/S0895-4356\(01\)00362-6](https://doi.org/10.1016/S0895-4356(01)00362-6)
 57. Horwitz RI, Yu EC (1985) Problems and proposals for interview data in epidemiological research. *Int J Epidemiol* 3:463–467. doi:[10.1093/ije/14.3.463](https://doi.org/10.1093/ije/14.3.463)
 58. Brizzolara S, Grandinetti A, Mor J (2002) Percentage of hysterectomy for pelvic organ prolapse in five ethnic groups. *Int Urogynecol J Pelvic Floor Dysfunct* 6:372–376. doi:[10.1007/s001920200082](https://doi.org/10.1007/s001920200082)
 59. Brett KM, Higgins JA (2003) Hysterectomy prevalence by Hispanic ethnicity: evidence from a national survey. *Am J Public Health* 2:307–312